

Breast Ultrasound Lesion Segmentation: Research Write-up

Version 1.0.0 · Site-only research report (not a journal submission)

Author: Surabhi Fadnavis (high-school senior, Georgia)

Live demo: <https://surabhif.github.io/Breast-ultrasound-lesion-segmentation/>

Repository: <https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>

Research demo — not for clinical use. Not clinician-reviewed. Not for diagnosis, screening, or care decisions.

1. Abstract

This high-school research project trains a ResNet-18 U-Net with an auxiliary benign-vs-malignant head on the BUSI breast ultrasound dataset, exports an INT8 ONNX model for fully in-browser inference, and reports honest internal and external metrics. On the held-out BUSI test set the served INT8 model reaches Dice 0.697 (lesion-only 0.764) and ROC-AUC 0.931. External validation on BUS-BRA and BrEaST keeps segmentation closer to internal Dice (0.714 / 0.627) while classification AUC drops (0.638 / 0.721). Cleaning, leakage, caliper-inpainting, and TTA uncertainty experiments are included. This write-up is published on the project site only.

2. Introduction

Breast ultrasound helps characterize masses, especially in dense breasts. Public datasets enable student projects but hide pitfalls: near-duplicate frames, burned-in calipers, and missing patient IDs. This project asks whether a small model can outline lesions in the browser — and how honest the scores remain under those pitfalls.

Related educational pages on the site cover BI-RADS context (a model score is not a BI-RADS category) and a surgeon's-view framing of size and margins (imaging size is not pathologic T-stage; model outlines are not surgical margins).

3. Data

3.1 BUSI (internal)

BUSI (Al-Dhabyani et al., Data in Brief, 2020) provides ~780 PNG ultrasound images labeled benign, malignant, or normal with masks. Multi-mask lesions are OR-merged. BUSI publishes no patient IDs; we group perceptual-hash near-duplicates so groups never cross splits. Annotation-flag rate (heuristic): 46.9% (366 / 780).

3.2 External sets

Primary external sets follow the pre-registered protocol in docs/EXTERNAL_VALIDATION_PROTOCOL.md: BUS-BRA (Gómez-Flores et al., 2024; Zenodo CC BY 4.0; n=1875, patients≈1064) and BrEaST (Pawłowska et al., 2024; TCIA CC BY 4.0; n=256, patients=256).

BUS-UCLM was not included because its Mendeley Data download requires a login (the request returned HTTP 403).

4. Methods

Architecture: ResNet-18 ImageNet encoder + slim U-Net decoder (160×160) with an auxiliary classification head. Training uses grouped stratified splits, paired augmentations, checkpointing on lesion Dice, and val-tuned segmentation threshold 0.4 with min-component area 40. The served artifact is INT8 ONNX models/v1.0.0/busi_unet.onnx (sha256 0bbf529d...). The browser runs onnxruntime-web (WASM) in a Web Worker with the same morphology post-process as the Python evaluation. Phase 2 added Telea

caliper-inpaint experiments, offline TTA uncertainty, and Demo threshold sliders.

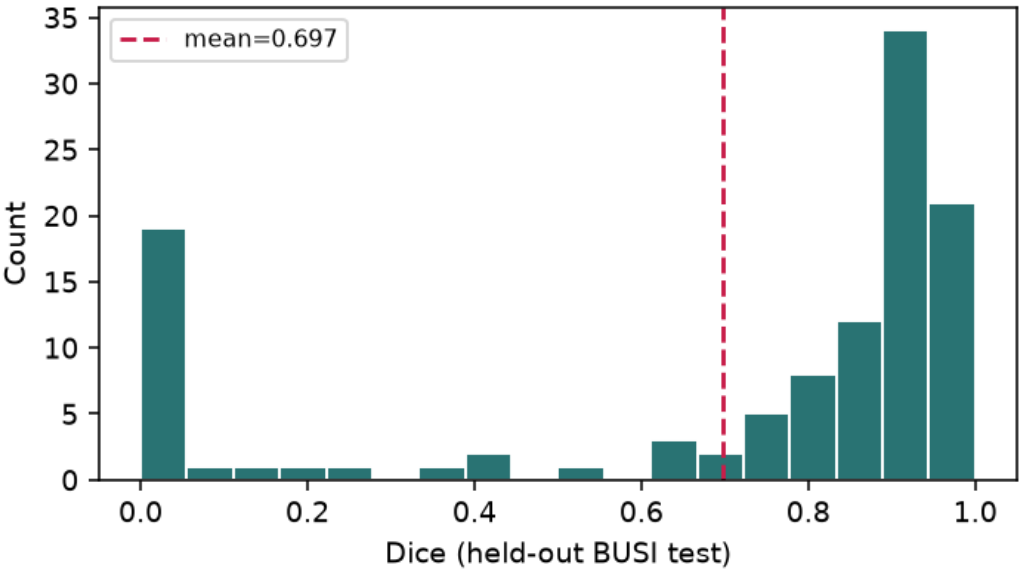
5. Results

5.1 Internal (BUSI held-out test)

Metric	FP32 training	Served INT8
Dice (all)	0.686	0.697
Lesion Dice	0.751	0.764
IoU	0.618	0.630
ROC-AUC	0.939	0.931
Sensitivity / Specificity	0.867 / 0.873	0.867 / 0.841
ECE	0.111	0.111
Normal FP masks	12/19	12/19

Sources: results/full_run.json, results/served_int8_test.json.

Figure 1. Served INT8 Dice distribution



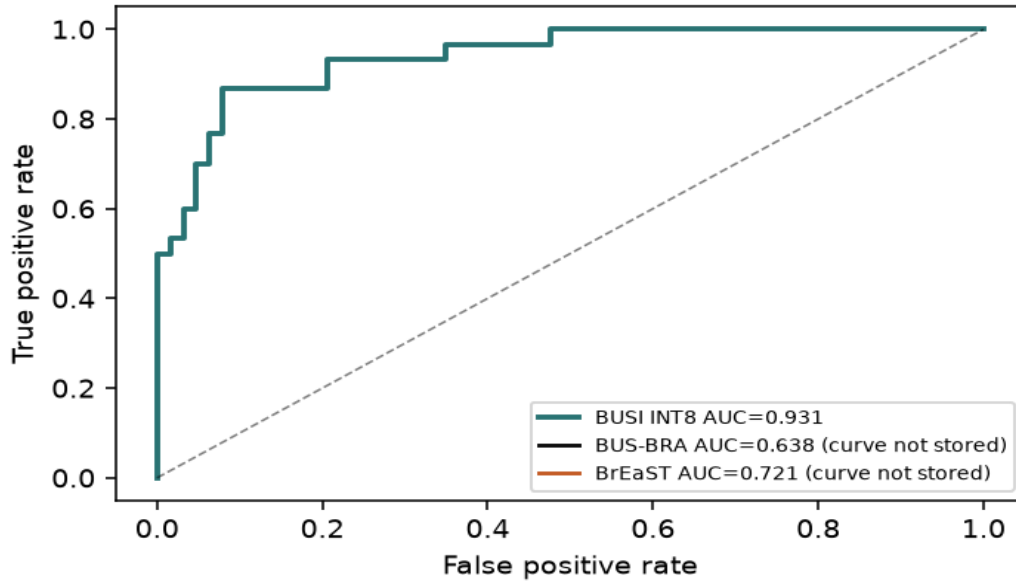
Per-image Dice on the held-out BUSI test set for the served INT8 model.

5.2 External validation

Set	Dice	Lesion Dice	AUC
BUSI internal	0.697	0.764	0.931
BUS-BRA	0.714	0.714	0.638
BrEaST	0.627	0.629	0.721

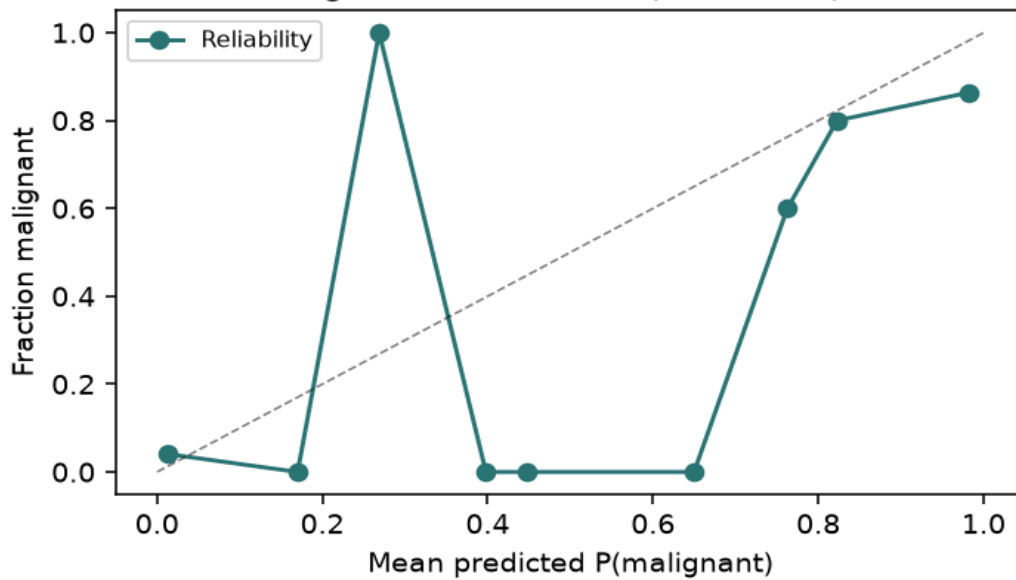
Sources: results/external/busbra.json, results/external/breast.json.

Figure 2. Classification ROC (internal vs external)



Classification ROC on BUSI versus external sets (when curves are stored).

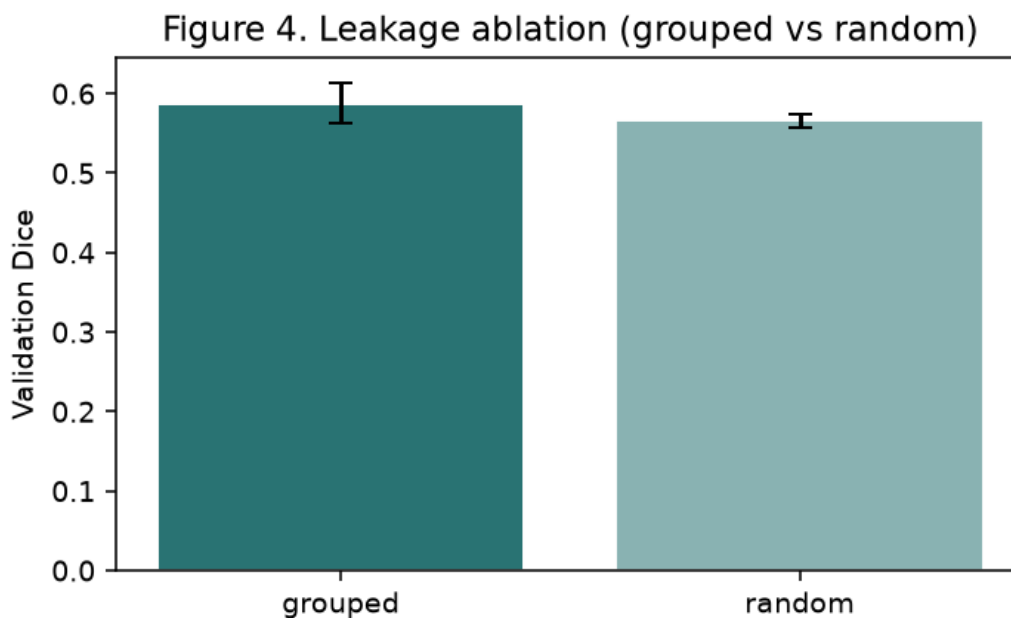
Figure 3. Calibration (BUSI INT8)



Reliability diagram for the served INT8 classifier (ECE 0.111).

5.3 Leakage ablation

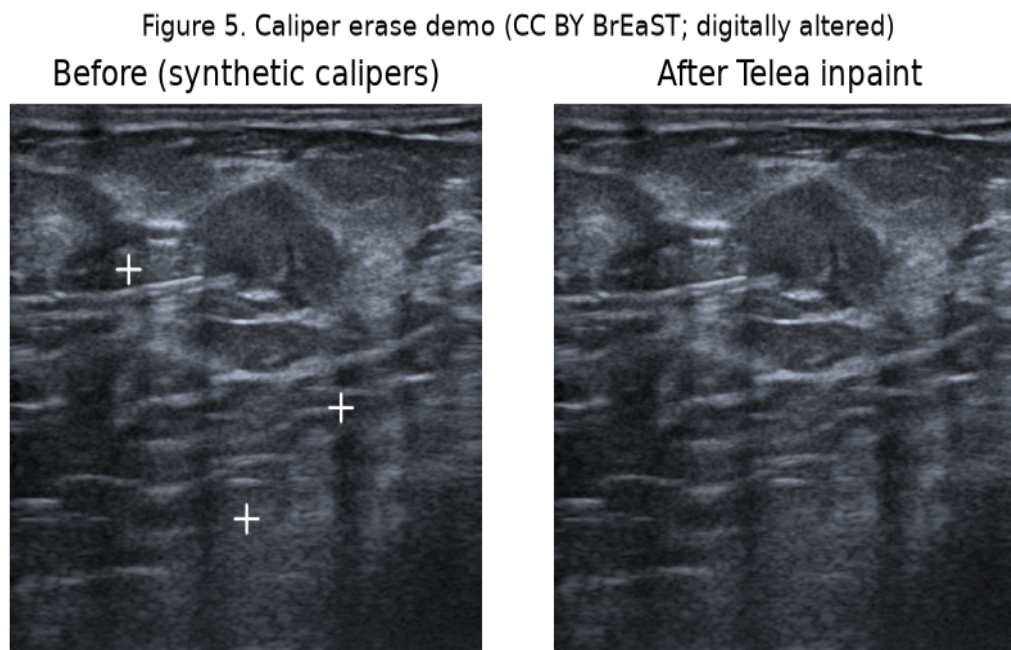
Short schedule (6 epochs × seeds 42,43,44), same grouped test: random vs grouped splits did not inflate validation Dice on this run (mean Δ val Dice random–grouped ≈ -0.022). Residual leakage risk remains because patient IDs are absent. Source: results/leakage_ablation.json.



Grouped versus random split validation Dice under the short ablation schedule.

5.4 Caliper / inpaint experiment

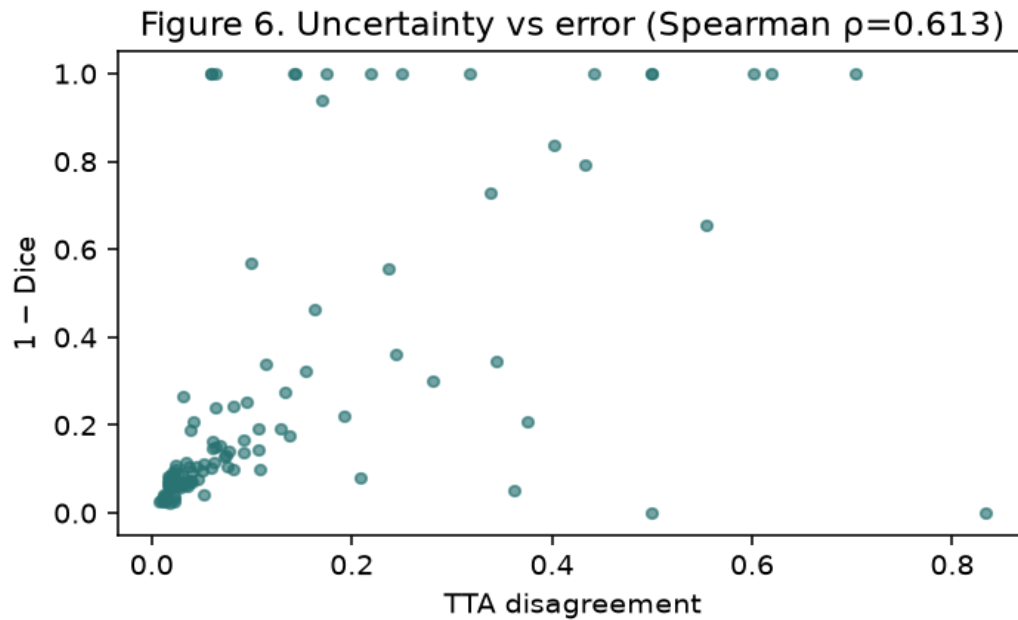
Flagged vs clean Dice gap on the original test: flagged 0.764 vs clean 0.628. Erasing markers (E-a) moved overall Dice from 0.697 to 0.702 and did not collapse flagged Dice toward clean. The best E-c retrain seed did not replace served v1.0.0 under the pre-registered swap rule: external Dice was BUS-BRA 0.626 (vs v1 0.714) and BrEaST 0.608 (vs v1 0.627); clean Dice 0.626; AUC 0.945. Source: results/inpaint_experiment.json.



Illustrative CC BY BrEaST frame with synthetic calipers removed by Telea inpainting.

5.5 Uncertainty (TTA)

Spearman ρ between TTA disagreement and $1 - \text{Dice} \approx 0.613$ on BUSI test ($n=112$). Risk-coverage at 80% keep $\approx \text{Dice } 0.778$. Interpretation: agreement under flips — not a calibrated error probability. Source: results/uncertainty.json.



Offline TTA disagreement versus segmentation error on the BUSI test set.

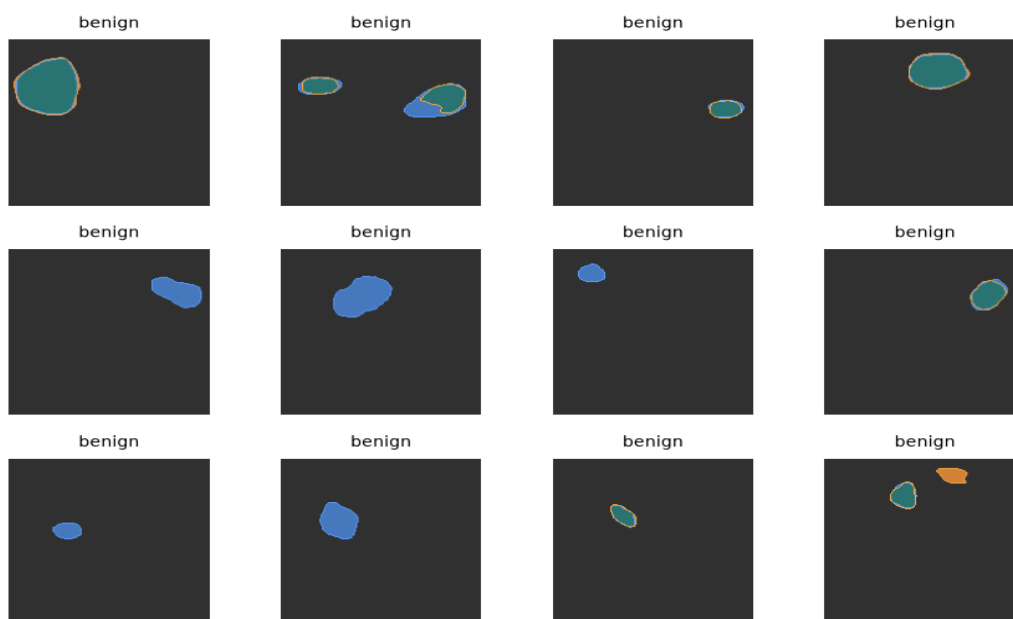
5.6 Measurement agreement (BrEaST)

Model vs expert longest diameter (mm) Pearson proxy ≈ 0.697 (n=252); area Pearson proxy ≈ 0.763 ; ≥ 20 mm threshold discordance $\approx 17.1\%$. Research only — not pathologic T-stage. Source: results/measurement_agreement.json.

5.7 Failure cases

Hard cases from the held-out test are published as outline-only silhouettes on the Model Errors explorer (BUSI ultrasound pixels withheld on new visuals). The grid below samples those outlines.

Figure 7. Failure-case outline silhouettes (BUSI; no ultrasound pixels)



Outline-only silhouettes for selected BUSI model-error cases.

6. Limitations

- No patient IDs → residual near-duplicate / patient leakage risk
- Domain shift: classification AUC drops sharply on externals
- Calipers / HUD may still act as shortcuts
- 160² resolution loses fine detail; normal-image false positives (12/19)
- Not clinically validated; no clinician review on this project
- A model score is not a BI-RADS category; imaging size is not surgical staging

7. Ethics and licensing

Code is MIT. Full BUSI is not redistributed (license unclear); cite Al-Dhabyani et al. 2020. External demo imagery uses CC BY sets with attribution. Site social preview uses CC BY external pixels (not BUSI). Research demo only. This site uses no analytics or cookies; Demo images are processed in the browser only.

8. AI-assistance statement

Project by Surabhi Fadnavis. The code, analysis, text, and video were produced with AI tools (Cursor).

9. References

1. Al-Dhabyani W, et al. Dataset of breast ultrasound images. Data in Brief. 2020;28:104863.

2. Mendelson EB, et al. ACR BI-RADS® Ultrasound. In: ACR BI-RADS® Atlas. Reston, VA: ACR; 2013.
3. Gómez-Flores W, et al. BUS-BRA breast ultrasound dataset. Zenodo; 2024. CC BY 4.0.
4. Pawłowska A, et al. BrEaST — Breast Cancer Dataset. TCIA; 2024. CC BY 4.0.
5. Moran MS, et al. SSO-ASTRO consensus guideline on margins for breast-conserving surgery. 2014.
6. Project repository: <https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>

Numbers in this PDF are auto-filled from committed results JSON by scripts/build_report_pdf.py. A CI check fails if PDF figures drift from those JSON files.